

Making China AIGC Compliance Evidence Machine-Readable: A Governance Pipeline Prototype for Synthetic Actor and Digital Human Platforms

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Technical paper draft based on a research prototype. This manuscript is not a formal publication and does not claim legal compliance certification.

Abstract

This paper presents a governance pipeline prototype for making China-facing AIGC compliance evidence machine-readable in synthetic actor and digital human platform scenarios. As generative AI is increasingly used in advertising, livestreaming, synthetic voice, virtual presenters, and digital human production, compliance evidence is no longer limited to general policy statements or final review reports. A single generated output may depend on actor authorization, consent scope, permitted usage region, model filing metadata, visible and machine-readable AIGC labeling, content safety review, complaint handling, and post-release audit records. In practice, these records are often fragmented across documents, platform logs, spreadsheets, review workflows, and manual approvals. To address this evidence gap, this project develops a synthetic governance pipeline prototype that maps selected China-facing AIGC governance themes into technical controls and machine-readable artifacts. The v0.3 implementation follows the structure: Policy/Controls/Evidence/Label Metadata/Provenance/Audit API. It includes policy-to-control mapping, request-level control evaluation, JSON Schema-validated evidence bundles, machine-readable label metadata, provenance manifests, deterministic evidence hashing, and a FastAPI-based Audit API for evidence retrieval and bundle verification. A seeded run generates 1,000 synthetic AIGC generation requests, producing 800 pass, 120 review, and 80 fail outcomes. The pipeline also generates 1,000 label metadata records, 1,000 provenance manifests, and 88 selected per-request evidence bundles. The automated test suite passes 22 tests, and an API smoke check confirms that a real generated evidence bundle returns `schema_valid: true` and `hash_valid: true`. The prototype does not process real personal data, biometric data, face data, voice data, contracts, company records, customer records, or production media. It does not provide legal advice or certify compliance with any law or standard. Its contribution is technical: it demonstrates how selected AIGC governance requirements can be translated into structured, queryable, schema-validated, and audit-ready evidence artifacts.

Keywords: AI governance, AIGC compliance, machine-readable evidence, digital human, synthetic actor, provenance, audit API, China AI regulation

1 Introduction

The rapid adoption of generative AI in advertising, livestreaming, digital human production, synthetic voice, and virtual actor services has made compliance evidence increasingly difficult to manage through conventional documentation alone. In China-facing AIGC platforms, a single generated output may depend on multiple compliance-relevant records: actor or asset authorization, consent scope, permitted usage region, model filing metadata, AIGC labeling, content safety review, complaint handling, and post-release monitoring. These records are often distributed across policy documents, review logs, content management systems, legal files, and manual approval workflows. Even when evidence exists, it may not

be structured in a way that another system, auditor, platform operator, or regulator-facing review process can query and verify.

This creates a practical gap between regulatory expectations and technical implementation. AI governance discussions increasingly emphasize transparency, accountability, traceability, and auditability, but many compliance workflows still rely on static reports or human-readable checklists. Such artifacts are useful for explanation, but they are limited as operational evidence. They cannot easily support automated verification, request-level evidence retrieval, provenance reconstruction, or consistent review across large volumes of generated content. In high-volume AIGC environments, this limitation becomes more visible: governance teams need not only to describe compliance controls, but also to produce evidence that is machine-readable, schema-validated, and reusable across audit workflows.

Recent research on AI-generated content transparency has moved in this direction. Work on EU AI Act Article 50 argues that transparency obligations should not be reduced to post-hoc labels added after content generation. Instead, transparency should be treated as a system architecture requirement, with outputs designed to support both human-understandable disclosure and machine-readable verification [1]. Similarly, emerging work on governance pipelines argues that AI governance should be embedded into the software lifecycle through pre-deployment validation, deployment governance, runtime monitoring, and post-incident audit mechanisms [2]. These developments suggest a broader shift in AI governance research: from high-level principles and compliance statements toward technical infrastructure that can produce durable and auditable artifacts.

This paper applies that shift to the China-facing AIGC context. It presents a synthetic governance pipeline prototype for AI-generated actor and digital human platforms. The prototype does not attempt to provide legal advice or certify real-world compliance. Instead, it explores how selected governance expectations around consent, usage scope, model filing, AIGC labeling, content safety, and complaint handling can be translated into machine-readable technical artifacts. The v0.3 implementation follows a pipeline structure: Policy/Controls/Evidence/Label Metadata/Provenance/Audit API. This structure is designed to show how policy obligations can be mapped to controls, how controls can be evaluated at request level, and how the resulting evidence can be validated, queried, and reviewed through technical interfaces.

The contribution of this paper is threefold. First, it develops a policy-to-control mapping that links selected China-facing AIGC governance themes to control checks, evidence fields, and generated artifacts. Second, it implements machine-readable compliance artifacts, including JSON Schema-validated evidence bundles, label metadata records, provenance manifests, and deterministic evidence hashes. Third, it introduces an audit-oriented API that supports request-level evidence lookup, label metadata retrieval, provenance queries, control exception review, risk and remediation lookup, and evidence bundle verification. Together, these components demonstrate a practical method for moving from static compliance reports toward machine-readable governance evidence.

The prototype is intentionally limited in scope. All records are synthetic, and the project does not process real personal data, biometric data, face data, voice data, contracts, company records, customer records, or production media. It does not implement production-grade authentication, secure evidence retention, cryptographic watermarking, regulator-facing filing integrations, or legally validated compliance workflows. The goal is narrower: to demonstrate the functional completeness and reproducibility of a machine-readable compliance evidence pipeline for research, portfolio development, and further system design.

2 Related Work

2.1 From AI Governance Principles to Operational Evidence

AI governance frameworks commonly emphasize transparency, accountability, robustness, safety, human oversight, fairness, and redress [10, 11]. These principles have shaped both international policy instruments and organizational AI governance practices. However, a recurring limitation is that principles often remain difficult to operationalize. A platform may state that it follows transparency or accountability requirements, but the actual evidence may be dispersed across documentation, review logs, system outputs, and manual approvals. This creates a gap between governance language and verifiable implementation.

This gap is especially visible in generative AI systems. AIGC services produce outputs at scale, and each output may involve multiple compliance-relevant decisions. In actor-like digital human scenarios, the relevant evidence is not limited to content safety. It also includes authorization, usage scope, territorial restrictions, model filing records, synthetic content labeling, complaint handling, and post-release traceability. These requirements are difficult to manage through static reports alone. Governance evidence needs to be structured at the level of requests and outputs, rather than only summarized at the organizational level.

Compliance-as-Code and machine-readable governance approaches respond to this problem by translating policy requirements into structured rules, controls, and evidence artifacts. Rather than treating compliance as a final document, these approaches treat it as part of the system workflow. The value of this approach is not that it eliminates human judgment. Instead, it makes the basis of human judgment easier to inspect, reproduce, and audit.

2.2 China-Facing AIGC Governance Context

China’s AIGC governance framework includes multiple layers of rules and technical guidance. The Interim Measures for the Management of Generative Artificial Intelligence Services address generative AI service governance, including content safety and service provider responsibilities [4]. The Provisions on the Administration of Deep Synthesis Internet Information Services address deep synthesis service governance, including provider obligations and content labeling expectations [5]. The Measures for the Labeling of Artificial Intelligence-Generated Synthetic Content and GB 45438-2025 further formalize expectations around identifying AI-generated synthetic content [6, 7]. The Personal Information Protection Law also remains relevant where AIGC systems involve personal information, biometric data, or identifiable likeness and voice data [9].

This regulatory context matters for synthetic actor and digital human platforms. In these scenarios, compliance evidence is not limited to whether an output is safe or labeled. A platform may need to record whether a virtual actor asset has authorization, whether the requested usage falls within the consent scope, whether the region of use is permitted, whether the generation model has filing or registration metadata, whether the output carries visible and machine-readable labels, whether content safety checks were completed, and whether complaints were handled within a defined process. These relationships are operational rather than purely textual. They require structured evidence that can connect requests, outputs, assets, models, labels, safety reviews, and complaint records.

The prototype in this paper uses these China-facing governance themes as background context. It does not provide a complete legal interpretation or claim that the implemented controls fully represent any regulation or standard. Instead, it uses selected obligations as a design basis for exploring how compliance expectations can be translated into machine-readable technical artifacts.

2.3 Machine-Readable Transparency and AIGC Labeling

Transparency in AIGC systems is often discussed through labeling. Visible labels help users understand that content was generated or modified by AI. However, visible labels alone are limited. They may be removed, edited, cropped, compressed, ignored by downstream platforms, or interpreted differently by different users. For audit purposes, a label also needs to be linked to a specific output, model, request, and verification record.

Recent work on AI-generated content transparency argues that transparency should not be treated as a post-hoc label attached after generation. A more robust approach is to design transparency into the system architecture. This means that generated outputs should support both human-understandable disclosure and machine-readable verification [1]. In practice, this requires metadata structures, stable identifiers, provenance records, validation mechanisms, and verification interfaces.

This project applies that idea to China-facing AIGC evidence. The prototype does not implement production watermarking or cryptographic content credentials. Instead, it models machine-readable labeling as structured metadata. Each generated output receives a label metadata record containing output ID, request ID, actor asset ID, model ID, content type, disclosure information, machine-readable format, verification status, and deterministic hashes. This design allows labeling to function as audit evidence, not merely as a user-facing notice.

2.4 Provenance and Auditability

Auditability requires more than a final pass or fail status. A reviewer often needs to reconstruct how a decision was reached, which records supported it, and whether the relevant checks occurred before release. Provenance is therefore central to machine-readable compliance evidence. It connects an output to the system events and records that produced it.

In AIGC actor and digital human contexts, provenance should link at least five types of information: the generation request, the actor or asset authorization, the model used, the labeling metadata, and the content safety decision. If a complaint is filed, complaint handling should also become part of the evidence chain. Without this linkage, the compliance status of an output is difficult to verify after publication.

The v0.3 prototype exports provenance manifests for each generated output. These manifests do not claim to represent production-grade secure provenance. They serve a narrower function: demonstrating how a synthetic platform can connect request-level events into a structured audit trail. This makes the project closer to an evidence infrastructure than a report generator.

2.5 Governance Pipelines

Governance pipeline research argues that AI governance should be embedded across the AI lifecycle rather than performed only through separate documentation. A typical governance pipeline may include pre-deployment validation, deployment governance, runtime monitoring, and post-incident governance. Each stage creates artifacts that can later be reviewed or audited [2]. This idea is also consistent with MLOps research, which emphasizes that model development and deployment should be supported by metadata stores, registries, testing, monitoring, and reproducible workflows [12].

This perspective is useful for AIGC compliance because generated content systems operate continuously. Compliance cannot be checked only once before deployment. Actor consent can expire, usage scope can be exceeded, model filing information can become outdated, label metadata can be missing, content safety checks can require escalation, and complaints can become overdue. A pipeline approach makes it possible to model these issues as control events and evidence records.

The prototype presented in this paper adapts the governance pipeline idea to a China-facing AIGC setting. It does not fully implement a production MLOps pipeline. Instead, it builds a smaller evidence-

oriented pipeline: Policy/Controls/Evidence/Label Metadata/Provenance/Audit API. This structure is intended to show how governance obligations can be transformed into request-level evidence and exposed through machine-readable interfaces.

3 Methodology

This project adopts a design-oriented methodology. Rather than conducting legal interpretation or evaluating a deployed production system, it builds a synthetic technical artifact that operationalizes selected AIGC governance requirements as machine-readable evidence. The methodology follows seven stages: synthetic operational log generation, policy-to-control mapping, request-level control evaluation, schema-validated evidence bundle construction, machine-readable label metadata generation, provenance manifest export, and Audit API verification.

3.1 Synthetic Operational Log Generation

The prototype simulates a China-facing AIGC actor and digital human advertising platform. It generates synthetic virtual actor assets, consent records, model filing records, generation requests, output records, labeling checks, content safety checks, and complaint records. Each generated request represents a content production event in which a client requests an AIGC output using a synthetic actor or digital human asset. The dataset is deterministic under a fixed seed, which supports reproducibility and regression testing. The synthetic design is intentional: it allows the prototype to model compliance-relevant relationships without using real personal data, biometric data, media files, contracts, or company records.

The generated records are operational logs only. They model relationships that would matter in a platform context, such as whether a generation request is linked to an active consent record, whether the requested usage is within scope, whether a model has filing metadata, whether an output has label metadata, whether content safety checks were completed, and whether a complaint is open or resolved. This makes the dataset suitable for evaluating evidence structures while avoiding privacy and data protection risks associated with real media or personal information.

3.2 Policy-to-Control Mapping

The project defines a structured mapping between selected policy sources, obligation categories, control identifiers, required evidence fields, generated artifacts, and pipeline stages. This mapping is used to connect abstract governance expectations to concrete technical checks. For example, AIGC labeling obligations are mapped to visible label and machine-readable label controls; consent-related obligations are mapped to active authorization, usage scope, and region scope checks; complaint handling expectations are mapped to synthetic SLA review and escalation controls.

This layer is important because machine-readable evidence is not only a data export problem. It also requires traceability between a policy expectation, a control decision, and the evidence field that supports that decision. Without this mapping, an evidence bundle may contain structured data but still fail to explain what governance obligation the data is intended to support.

Table 1: Policy-to-Control Mapping

Governance theme	Control examples	Required evidence	Pipeline stage
Actor authorization	Active consent exists	Actor asset ID, consent ID, consent status, validity period	Pre-release validation
Usage scope	Requested usage within consent scope	Requested usage, allowed usage, consent ID	Pre-release validation
Region scope	Requested region within consent scope	Requested region, allowed regions, consent ID	Pre-release validation
Model filing metadata	Model filing present and registered	Model ID, filing status, registration reference	Deployment governance
Visible AIGC label	Visible label present or review required	Output ID, visible label status, disclosure text	Output transparency
Machine-readable label	Metadata label present and verifiable	Label ID, output hash, metadata hash, verification status	Output transparency
Content safety review	Risk level matched with appropriate action	Safety check ID, risk level, action, categories	Pre-release validation
Complaint handling	Complaint handled within synthetic SLA	Complaint ID, status, received time, resolution time	Post-release governance
Provenance export	Output linked to request, model, label, safety, complaint	Provenance ID, event list, evidence references	Audit preparation
Audit verification	Evidence bundle schema and hash valid	Schema status, hash status, case reference	Post-incident audit

3.3 Request-Level Control Evaluation

Each synthetic generation request is evaluated against the defined control catalog. The evaluator joins the request with the relevant actor consent, model filing record, output record, labeling check, content safety check, and complaint record where available. Each control returns one of three statuses: pass, review, or fail. A request fails if any control fails, is marked as review if no control fails but at least one control requires manual attention, and passes only if all controls pass.

This stage produces a structured assessment result for each request and allows the prototype to generate aggregate compliance summaries as well as request-level evidence. The request-level design is important because AIGC platforms may generate large volumes of outputs under different actors, models, campaigns, clients, and usage scenarios. Aggregate reports alone cannot show whether a specific output has valid evidence.

3.4 Schema-Validated Evidence Bundle Construction

The project builds both aggregate evidence bundles and selected per-request evidence bundles. The aggregate bundle summarizes the synthetic run, source inventory, control catalog, request results, policy mapping summary, artifact index, and audit summary. Each per-request bundle connects the case reference, control results, policy mappings, label metadata, provenance manifest, and audit metadata for a specific request.

JSON Schema validation is added to ensure that these evidence bundles are not merely informal JSON files but stable machine-readable structures. Each bundle includes audit information such as schema validity, deterministic evidence hash, generation metadata, retention class, and synthetic data notice. This makes the evidence easier to check programmatically and reduces ambiguity about required fields and expected structure.

3.5 Machine-Readable Label Metadata Generation

Earlier versions of the prototype represented AIGC labeling through visible and invisible label checks. The v0.3 pipeline extends this by generating a label metadata record for each output. Each record includes the output identifier, request identifier, actor asset identifier, model identifier, content type, visible disclosure text, machine-readable label format, AI-generated status, deterministic output hash, metadata hash, verifier URI, verification timestamp, and verification status.

In the current prototype, label verification uses a simplified rule: if the invisible or metadata label is present, the status is valid; if it is absent, the status is missing. This design is not intended to replicate production watermarking. It demonstrates how AIGC labeling can be represented as structured metadata that supports automated review.

3.6 Provenance Manifest Export

For each generated output, the prototype creates a provenance manifest that connects the request, actor asset, consent record, model, label metadata, safety check, complaint record, and final assessment result. The manifest includes ordered provenance events such as request creation, consent validation, model filing check, output generation, label attachment, content safety review, complaint check, and assessment completion.

The output is exported both as a JSON list and as NDJSON, allowing the provenance data to be consumed either by humans or downstream machine-processing workflows. This stage responds to a central audit need: compliance review often requires not only knowing whether a request passed, but also reconstructing how the output moved through the governance process.

3.7 Audit API Verification

The prototype provides a FastAPI-based interface that reads generated artifacts from the outputs directory and exposes them through queryable endpoints. The API supports health checks, request-level evidence retrieval, label metadata lookup, provenance lookup, control exception review, risk and remediation access, and evidence bundle verification. The verification endpoint checks whether a submitted evidence bundle is schema-valid, whether required identifiers and control results are present, and whether the evidence hash is consistent.

This API is a demonstration interface, not a production regulatory reporting system. Its purpose is to show how machine-readable compliance evidence can be queried and verified after generation.

Together, these seven stages form a governance pipeline rather than a static report generator. The pipeline begins with synthetic operational records, maps policy expectations into controls, evaluates those controls, packages evidence into validated structures, attaches label metadata, exports provenance, and exposes audit functions through an API. This design allows the project to test whether selected AIGC governance requirements can be represented as structured, reproducible, and machine-verifiable artifacts.

4 System Architecture

The v0.3 prototype is organized as a governance evidence pipeline. The architecture links policy sources, control evaluation, evidence generation, label metadata, provenance export, and audit verification. Its purpose is to show how selected compliance obligations can be transformed into structured artifacts that are readable by both humans and machines.

The first layer is policy-to-control mapping. This layer connects selected governance themes to specific control identifiers and evidence requirements. It does not attempt to provide a complete legal interpretation. Instead, it defines an operational mapping between policy expectations and technical

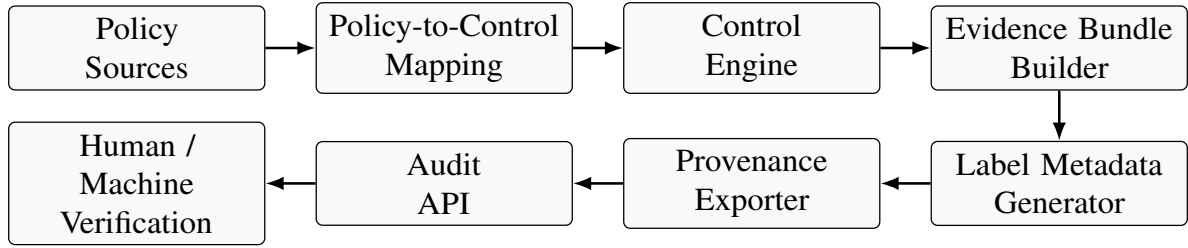


Figure 1: Governance evidence pipeline for machine-readable China-facing AIGC compliance evidence.

checks. For example, actor authorization is mapped to active consent and scope controls; AIGC labeling is mapped to visible label and machine-readable metadata controls; complaint handling is mapped to synthetic SLA and escalation checks.

The second layer is the control engine. It evaluates each generation request against the configured controls. The control engine joins records from the synthetic dataset, including consent records, model filing records, output records, labeling checks, safety checks, and complaint records. Each control returns pass, review, or fail. This produces both request-level and aggregate assessment results.

The third layer is the evidence bundle builder. It packages control results and supporting records into structured evidence artifacts. The aggregate bundle summarizes the full synthetic run, while per-request bundles provide more detailed evidence for selected requests. The v0.3 implementation adds schema validation and audit metadata to strengthen the machine-readability of these bundles.

The fourth layer is the label metadata generator. This layer generates one machine-readable label metadata record for each output. The record links the output to its request, actor asset, model, content type, disclosure information, verification status, and deterministic hashes. This supports automated label review and provides a clearer evidence structure than a simple visible/invisible label flag.

The fifth layer is the provenance exporter. It creates provenance manifests that connect each output to the governance events behind it. Each manifest includes identifiers for the request, actor asset, consent record, model, label metadata, safety check, complaint record where applicable, and final assessment. It also records ordered events such as request creation, consent validation, model filing check, output generation, label attachment, safety review, complaint check, and assessment completion.

The sixth layer is the Audit API. The API exposes generated artifacts through queryable endpoints. It supports request-level evidence lookup, output-level label metadata lookup, provenance retrieval, control exception review, risk register access, remediation item access, and evidence bundle verification. This layer demonstrates how machine-readable evidence can be used after generation for internal audit, platform review, or regulator-facing evidence retrieval.

The architecture intentionally separates evidence generation from legal certification. The system can show whether synthetic evidence records exist, whether they follow the expected schema, and whether they are linked across the pipeline. It cannot determine whether a real organization has satisfied legal obligations. This distinction is central to the prototype design.

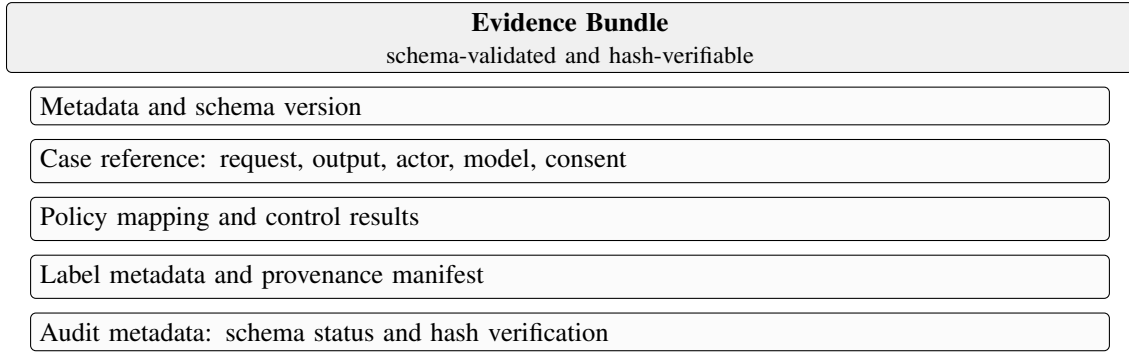


Figure 2: Internal structure of a schema-validated and hash-verifiable evidence bundle.

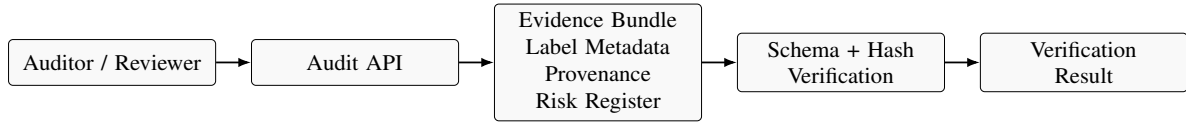


Figure 3: Prototype Audit API flow for evidence lookup and verification.

Table 2: Generated Machine-Readable Artifacts

Artifact	Format	Purpose
assessment_results.json	JSON	Stores request-level control results and aggregate assessment summary
risk_register.json	JSON	Groups open review and fail items into synthetic risk records
remediation_items.json	JSON	Provides proposed actions for unresolved synthetic risks
evidence_bundle.json	JSON	Stores aggregate compliance evidence for the full synthetic run
evidence_bundles/{request_id}.json	JSON	Stores selected per-request evidence chains
label_metadata.json	JSON	Stores machine-readable label metadata for generated outputs
label_verification_results.json	JSON	Stores label verification status for each output
provenance_manifest.json	JSON	Stores complete provenance manifests for generated outputs
provenance_export.ndjson	NDJSON	Provides machine-processing-friendly provenance export
request_control_matrix.csv	CSV	Provides compact request-control summary for review
evidence_report.md	Markdown	Provides human-readable report output

Table 3: Audit API Endpoints

Endpoint	Function
GET /health	Checks whether the API is running
GET /requests/{request_id}/evidence	Retrieves request-level evidence bundle
GET /outputs/{output_id}/label-metadata	Retrieves machine-readable label metadata for an output
GET /outputs/{output_id}/provenance	Retrieves provenance manifest for an output
GET /controls/{control_id}/exceptions	Lists review/fail exceptions for a selected control
GET /risks	Retrieves synthetic risk register
GET /remediations	Retrieves remediation items
POST /verify/evidence-bundle	Validates evidence bundle structure and hash consistency

5 Implementation

The prototype is implemented as a Python-based research artifact. Its purpose is to demonstrate how selected compliance themes can be represented through synthetic operational logs, request-level controls, structured evidence bundles, label metadata records, provenance manifests, and an audit-facing API.

The core implementation contains five functional layers. The synthetic data generation layer creates virtual actor assets, consent records, model filing records, generation requests, outputs, labeling checks, content safety checks, and complaint records. The policy engine evaluates each request against configured controls and returns pass, review, or fail results. The evidence builder packages these results into aggregate

and per-request evidence bundles. The label metadata and provenance modules extend the evidence model by generating output-level metadata and event-level provenance chains. The Audit API exposes these artifacts through queryable endpoints.

The v0.3 implementation adds a JSON Schema layer. Schemas are defined for evidence bundles, label metadata, provenance manifests, and audit events. This layer helps ensure that generated artifacts follow predictable structures and can be validated before being used in downstream workflows. In this sense, the prototype treats JSON not merely as an export format but as a structured evidence interface.

The label metadata module generates one machine-readable label record for each output. The metadata does not embed a real watermark into media content. Instead, it records the label's relationship to the output, request, actor asset, model, disclosure text, machine-readable format, verification status, and deterministic hashes. This approach is intentionally lightweight, but it captures the structure needed for later audit and automated review.

The provenance module generates one manifest per output. Each manifest contains a case reference and a sequence of governance-relevant events. These events include request creation, consent validation, model filing check, output generation, label attachment, content safety review, complaint check, and assessment completion. The manifest connects records that would otherwise remain separate.

The Audit API is implemented using FastAPI. It reads generated artifacts from the local output directory and supports evidence lookup and verification endpoints. The API is not designed for production deployment. It does not include authentication, authorization, secure retention, or regulator integration. It is included to demonstrate how machine-readable evidence can be exposed for review after generation.

The implementation is intentionally reproducible. The seeded command `python -m src.cli --seed 20260625` regenerates the synthetic dataset and associated artifacts. Automated tests verify schema validation, label metadata generation, provenance generation, deterministic hashing, Audit API endpoints, and seeded pipeline regression behavior.

6 Evaluation

The evaluation focuses on the functional completeness, reproducibility, and artifact consistency of the v0.3 prototype. It does not evaluate real-world legal compliance, production security, or regulatory acceptance. Because the project uses synthetic data only, the evaluation is designed to answer a narrower technical question: whether the pipeline can consistently generate, validate, link, and expose machine-readable compliance evidence across a realistic set of synthetic AIGC requests.

The seeded run was executed using the command:

```
python -m src.cli --seed 20260625
```

The pipeline generated 1,000 synthetic AIGC generation requests. The assessment results followed the intended demonstration distribution: 800 requests were marked as pass, 120 as review, and 80 as fail. In addition to the original assessment artifacts, the v0.3 pipeline generated 1,000 machine-readable label metadata records, 1,000 provenance manifests, and 88 selected per-request evidence bundles. These selected bundles include representative pass, review, and fail cases, as well as complaint-related cases.

Table 4: Seeded Run Validation Results

Metric	Result
Synthetic generation requests	1,000
Pass	800
Review	120
Fail	80
Label metadata records	1,000
Provenance manifests	1,000
Selected per-request evidence bundles	88
Automated tests	22 passed, 1 warning
Audit API smoke check	schema_valid=true, hash_valid=true

The generated artifact set includes both human-readable and machine-readable outputs. Human-readable outputs include Markdown reports and tabular summaries. Machine-readable outputs include assessment results, risk register, remediation items, evidence bundles, label metadata records, label verification results, provenance manifests, and NDJSON provenance export. This distinction matters because the project is designed to support both human review and automated verification. A compliance reviewer can inspect the report and selected bundles, while a machine process can validate schema conformance, query request-level evidence, and verify hash consistency.

The schema validation layer provides a basic integrity check for the generated evidence structures. Evidence bundles, label metadata records, provenance manifests, and audit events are validated against JSON Schema definitions. This reduces ambiguity in the artifact format and makes the outputs more suitable for downstream review or API-based retrieval. In the current prototype, schema validation is used to confirm that generated records contain required identifiers, case references, control results, label metadata, provenance information, and audit metadata.

The machine-readable label metadata output was also evaluated for completeness. The seeded run produced one label metadata record for each generated output. Each record links the output to its request, actor asset, model, content type, disclosure fields, machine-readable format, verification status, and deterministic hashes. This does not prove that a real watermark or production metadata label would survive downstream editing or format conversion. It does, however, demonstrate that the project can represent AIGC labeling as queryable evidence rather than as a simple yes/no field.

The provenance layer was evaluated by checking that each generated output had a corresponding provenance manifest. The seeded run produced 1,000 provenance manifests. Each manifest connects the request, consent, model, output, label, safety review, complaint status where applicable, and final assessment. The provenance events allow the pipeline to reconstruct the governance path of a generated output. This is useful for audit scenarios in which the reviewer needs to understand not only the final compliance status but also which evidence records supported that status.

The Audit API was evaluated through a smoke check. The health endpoint returned normally, and the evidence bundle verification endpoint returned `schema_valid: true` and `hash_valid: true` for a real generated bundle. This confirms that the API can read generated artifacts, validate evidence structure, and verify deterministic hash consistency under the prototype setting. The API also supports request-level evidence lookup, output-level label metadata lookup, provenance lookup, control exception review, and access to risk and remediation records.

The automated test suite passed 22 tests, with one deprecation warning from the FastAPI/Starlette TestClient dependency. The warning does not affect prototype functionality. The tests cover schema validation, label metadata generation, provenance manifest generation, deterministic hash behavior, Audit API health and lookup endpoints, and regression checks for the seeded pipeline output.

Overall, the evaluation shows that the v0.3 prototype can reproducibly generate a complete set of synthetic machine-readable compliance artifacts. The results support the project's core claim at

the prototype level: selected AIGC governance requirements can be translated into structured controls, validated evidence bundles, label metadata, provenance records, and audit-facing API endpoints. However, the evaluation should not be read as evidence of real-world compliance. The prototype does not use production data, does not implement production security controls, and has not been reviewed by legal or regulatory experts. Its value is as a technical demonstration of how compliance evidence could be structured and exposed within an AI governance pipeline.

7 Discussion

The prototype shows that AIGC compliance evidence can be treated as a technical artifact rather than only as a written report. This does not remove the need for legal judgment or human review. However, it changes the form of evidence available to those reviewers. Instead of asking whether a platform has a general compliance policy, the reviewer can inspect whether a specific request has linked consent, model filing, label metadata, safety review, complaint handling, and provenance records.

This shift matters because AIGC platforms operate at high volume. Static compliance reports can describe governance arrangements, but they are less effective for request-level verification. A machine-readable evidence pipeline allows evidence to be retrieved by request ID, output ID, control ID, or risk category. This makes audit workflows more scalable and less dependent on manual file reconstruction.

The label metadata layer is especially important. AIGC transparency is often reduced to whether a user can see a label. That is necessary but insufficient. For platform governance, the more important question is whether the label can be associated with a specific output and verified later. By generating label metadata records, the prototype makes labeling part of the evidence chain. This is closer to the idea of transparency as architecture.

The provenance layer adds another form of audit value. A final pass or fail status is not enough to reconstruct compliance history. Provenance manifests allow the reviewer to see which events occurred and which records supported the assessment. In a real platform, this type of structure could support incident review, complaint investigation, client due diligence, and regulator-facing evidence retrieval.

The Audit API further changes how evidence can be used. Without an API, evidence remains a set of generated files. With an API, evidence becomes queryable and verifiable. A user can retrieve request-level evidence, inspect label metadata, check provenance, review control exceptions, and validate a submitted bundle. The current implementation is intentionally simple, but it demonstrates the core mechanism needed for audit-ready evidence infrastructure.

At the same time, the prototype also shows the limits of engineering approaches to compliance. A schema can validate whether a field exists, but it cannot determine whether the legal interpretation behind that field is correct. A deterministic hash can detect whether an evidence bundle has changed, but it does not guarantee that the original evidence was legally sufficient. A provenance manifest can record synthetic events, but it cannot prove the integrity of a production system without secure logging, access control, retention policies, and independent validation.

The project should therefore be understood as a bridge between AI governance theory and technical implementation. It does not replace legal analysis. It demonstrates how legal and governance expectations can be translated into structured, machine-readable artifacts that support later review. This makes the prototype useful for research and portfolio development, while also showing what additional work would be required for production deployment.

8 Limitations and Future Work

This project has several limitations. First, the dataset is entirely synthetic. The prototype does not process real personal data, biometric data, face data, voice data, contracts, company records, customer records, or

production media. This design is appropriate for a research prototype, but it means the evaluation cannot demonstrate real-world compliance performance.

Second, the policy mapping is illustrative. It is informed by selected China-facing AIGC governance themes, but it is not a complete legal interpretation. The project does not provide legal advice, does not certify compliance, and does not determine whether a real AIGC platform satisfies any law, regulation, standard, or filing obligation.

Third, the label metadata layer is simplified. The prototype does not implement cryptographic watermarking, production metadata embedding, content credentials, or robustness testing under editing, compression, cropping, format conversion, or downstream platform processing. The label metadata records demonstrate structure and verifiability, not production-grade persistence.

Fourth, the provenance layer is not secured. The prototype creates provenance manifests from synthetic records, but it does not implement immutable logging, secure timestamping, access control, tamper-resistant storage, or external attestation. These would be necessary for production audit use.

Fifth, the Audit API is a demonstration interface. It does not include authentication, authorization, rate limiting, secure retention, regulator-facing filing integration, or enterprise workflow management. It is useful for showing how evidence can be queried and verified, but it is not designed as a production regulatory reporting system.

Future work can extend the prototype in several directions. The first direction is stronger policy mapping, including more detailed legal review and clearer distinction between legal obligations, technical controls, and organizational procedures. The second direction is stronger evidence integrity, including signed evidence bundles, immutable logs, and secure retention policies. The third direction is label robustness, including metadata survival tests and compatibility with emerging content provenance standards. The fourth direction is runtime monitoring, where consent expiration, label loss, complaint escalation, and safety incidents are monitored continuously rather than simulated in batch output. The fifth direction is a dashboard or reviewer interface, which could help human auditors inspect risks, evidence chains, and remediation status more efficiently.

The current v0.3 prototype should therefore be viewed as a foundation. It demonstrates that selected AIGC compliance requirements can be transformed into machine-readable evidence artifacts, but it also leaves open the harder questions of legal validation, production security, and institutional adoption.

9 Conclusion

This paper presented a v0.3 governance pipeline prototype for making China-facing AIGC compliance evidence machine-readable in synthetic actor and digital human platform scenarios. The project began from a practical problem: AIGC compliance evidence is often available in fragmented, human-readable forms, but not structured in a way that supports automated verification, request-level audit, or provenance reconstruction. This limitation becomes more serious in high-volume content generation environments, where each output may involve authorization, model filing, labeling, safety review, and complaint handling records.

The prototype addresses this problem by organizing compliance evidence as a pipeline: Policy/Controls/Evidence/Label Metadata/Provenance/Audit API. This structure allows selected policy expectations to be mapped to technical controls, evaluated at request level, packaged into schema-validated evidence bundles, linked with machine-readable label metadata, connected through provenance manifests, and exposed through an audit-oriented API. The result is not merely a set of compliance reports, but a structured evidence infrastructure that supports both human review and machine verification.

The seeded evaluation shows that the pipeline can reproducibly generate 1,000 synthetic AIGC requests, classify them into pass, review, and fail outcomes, and produce the corresponding evidence artifacts. The v0.3 run generates 1,000 label metadata records, 1,000 provenance manifests, and 88

selected per-request evidence bundles. Schema validation, deterministic hashing, and API-level bundle verification provide basic integrity checks within the prototype environment.

The broader implication is that AI governance can be made more operational when compliance evidence is treated as part of system architecture rather than as a document produced after the fact. For AIGC platforms, this means that labels, consent records, model metadata, safety decisions, complaints, and audit trails should not be isolated records. They should be linked through stable identifiers, validated schemas, provenance structures, and queryable interfaces.

At the same time, the prototype remains limited. It uses only synthetic data and does not establish real-world legal compliance. It does not implement production-grade authentication, secure evidence retention, cryptographic watermarking, content credential standards, immutable logging, regulator-facing filing integrations, or legally validated workflows. These limitations are not incidental; they define the boundary between a research prototype and a production compliance system.

Future work should therefore focus on five areas: deeper policy mapping with legal review, stronger evidence integrity through signed bundles and tamper-resistant logs, more realistic label robustness testing, runtime monitoring for consent expiry and complaint escalation, and human-facing reviewer interfaces. The current prototype provides a foundation for these extensions by demonstrating that selected AIGC governance requirements can be translated into structured, reproducible, queryable, and audit-ready machine-readable evidence.

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